SIMATS ENGINEERING

ASSESSMENT TOOL 1 - Short Answer Test

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO1: Apply the fundamental concepts, image formation models, and processing
Techniques for analyzing Computer Vision systems. (BTL 3)
Assessment Tool Weightage: 40%

Code of Conduct:

I **K Sampath Reddy (192424285)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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Question 1: Real-World Scene Capture to Digital Image Representation

Question

Construct a comprehensive and well-organized concept map that accurately identifies all key concepts from literature involved in the pipeline from real-world scene capture to digital image representation, establishes clear and meaningful relationships among intermediate transformations and dependencies, integrates insights from multiple studies, and presents the information in a clear and visually structured manner.

1. Introduction

Computer Vision enables computers to acquire, process, analyze, and interpret visual information from the real world. Before a computer can recognize an object, detect faces, classify images, or perform scene understanding, the physical scene must first be transformed into a digital image. This transformation involves illumination, image formation, sensing, sampling, quantization, preprocessing, feature extraction, and digital representation.

2. Literature Review

- Gonzalez & Woods (2018): image acquisition and digital representation.
- Szeliski (2022): preprocessing and image formation.
- Forsyth & Ponce: geometric and photometric transformations.
- Goodfellow et al. (2016): importance of image representation for deep learning.
- OpenCV documentation: practical preprocessing and feature extraction.

3. Concept Map

```
REAL-WORLD SCENE
     |
Illumination + Object Surface
     |
Reflection
     |
Camera Lens
     |
```

Optical Image Formation
|
CCD/CMOS Sensor
|
Analog Signal
|
ADC
|
Sampling + Quantization
|
Digital Pixel Matrix
|
Preprocessing
|
Feature Extraction
|
Machine Learning / Deep Learning
|
Recognition / Decision

4. Flowchart

Real World Scene → Camera → Sensor → ADC → Sampling → Quantization → Digital Image → Preprocessing → Feature Extraction → Computer Vision Algorithm → Output

5. Detailed Explanation

The process starts with illumination, where light interacts with object surfaces and reflects into the camera lens. The optical system focuses this light onto a CCD or CMOS sensor that converts photons into electrical signals. These analog signals are digitized using an Analog-to-Digital Converter through sampling and quantization, creating a pixel matrix. The digital image is then preprocessed using filtering, enhancement and normalization before feature extraction identifies edges, textures, corners and shapes. Modern deep learning methods automatically learn hierarchical features and use them for classification, detection and segmentation. This pipeline forms the basis of applications including autonomous driving, medical imaging, surveillance and robotics. Each stage depends on the quality of the previous stage, making accurate acquisition and preprocessing essential.

6. Applications

- Autonomous driving
- Medical imaging
- Face recognition
- Industrial inspection
- Robotics
- Remote sensing
- Smart surveillance
- Augmented reality

7. Conclusion

The pipeline from a real-world scene to a digital image integrates optics, electronics, image processing and artificial intelligence. Understanding the relationships among illumination, sensing, digitization, preprocessing and feature extraction is fundamental to building accurate Computer Vision systems.

8. IEEE References

1. R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.
2. R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed., Springer, 2022.
3. D. A. Forsyth and J. Ponce, Computer Vision: A Modern Approach.
4. I. Goodfellow et al., Deep Learning, MIT Press, 2016.
5. R. Hartley and A. Zisserman, Multiple View Geometry in Computer Vision.
6. G. Bradski and A. Kaehler, Learning OpenCV.
7. OpenCV Documentation.
8. D. Marr, Vision.
9. J. Canny, IEEE TPAMI, 1986.
10. D. Lowe, IJCV, 2004.

Question 2: Hierarchical Computer Vision System

Question

Construct a hierarchically structured concept map that identifies major stages of visual data handling in a Computer Vision system, clearly illustrates logical relationships and contributions of each stage toward generating meaningful outputs, incorporates relevant literature connections, and ensures clarity and readability.

1. Introduction

Computer Vision enables computers to interpret images and videos through a hierarchy of processing stages. Visual information progresses from image acquisition to preprocessing, segmentation, feature extraction, model learning, and decision making, allowing systems to generate meaningful outputs for real-world applications.

2. Literature Review

- Szeliski (2022): Layered Computer Vision architecture.
- Forsyth & Ponce: Image preprocessing and feature extraction.
- Goodfellow et al. (2016): Hierarchical feature learning with deep learning.
- LeCun et al. (2015): Deep learning revolution in Computer Vision.
- OpenCV Documentation: Practical implementation of vision algorithms.

3. Hierarchical Concept Map

COMPUTER VISION SYSTEM
│
Image Acquisition
│
Image Preprocessing
│
Image Segmentation
│
Feature Extraction
│
Feature Representation
│
Machine Learning / Deep Learning

│
Classification / Detection / Recognition
│
Decision Making
│
Applications

4. Flowchart

Image Acquisition → Preprocessing → Segmentation → Feature Extraction → Feature Representation → ML/DL Model → Classification/Detection → Decision Making → Application Output

5. Detailed Explanation

A Computer Vision system starts with image acquisition using cameras or sensors. The captured image is preprocessed to remove noise, normalize intensity and improve quality. Segmentation separates important regions from the background. Feature extraction identifies edges, textures, shapes and colors, while modern deep learning models automatically learn hierarchical representations. Feature vectors are supplied to machine learning or deep learning models such as CNNs, ResNet, YOLO and Vision Transformers for classification, detection or segmentation. The interpreted information is then used for decision making in applications including autonomous driving, healthcare, surveillance, robotics and industrial inspection. Each stage depends on the successful completion of previous stages, making the hierarchical organization essential for reliable Computer Vision systems.

6. Applications

- Autonomous vehicles
- Medical imaging
- Facial recognition
- Smart surveillance
- Robotics
- Agriculture
- Industrial inspection
- Remote sensing
- Traffic monitoring
- AR/VR

7. Conclusion

The hierarchical Computer Vision pipeline transforms raw image data into semantic understanding through sequential processing stages. Advances in deep learning have improved accuracy and automation, making Computer Vision indispensable across many industries.

8. IEEE References

1. R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed., Springer, 2022.
2. D. Forsyth and J. Ponce, Computer Vision: A Modern Approach.
3. I. Goodfellow et al., Deep Learning, MIT Press, 2016.
4. R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.
5. Y. LeCun et al., Nature, 2015.
6. K. He et al., CVPR, 2016.
7. A. Krizhevsky et al., NeurIPS, 2012.
8. J. Redmon et al., CVPR, 2016.
9. O. Ronneberger et al., MICCAI, 2015.
10. OpenCV Documentation.

Question 3: Image Resolution and Intensity Levels

Question

Create a detailed concept map that accurately identifies concepts related to image resolution and intensity levels, establishes clear cause–effect relationships with image quality and usability, integrates supporting evidence from literature, and presents the concepts in a logically organized and visually clear format.

1. Introduction

Image resolution and intensity levels are fundamental properties of digital images. Spatial resolution determines the number of pixels representing an image, while intensity resolution (bit depth) determines the number of brightness levels available. Together they influence image quality, storage, processing speed and Computer Vision accuracy.

2. Literature Review

- Gonzalez & Woods (2018): spatial and intensity resolution.
- Szeliski (2022): resolution affects recognition accuracy.
- Pratt: importance of quantization.
- Forsyth & Ponce: role in segmentation.
- OpenCV: resizing and histogram equalization.

3. Concept Map

```
DIGITAL IMAGE
│
├─ Spatial Resolution (Pixels)
│     │
│     └→ Image Detail
│
├─ Intensity Resolution (Bit Depth)
│     │
│     └→ Brightness Precision
│
└→ Image Quality
     │
     ├→ Better Analysis
```

├→ Better Visualization
├→ Higher Accuracy
└→ Better Usability

4. Cause-Effect Flowchart

Higher Spatial Resolution → More Pixels → Greater Detail → Better Feature Detection → Higher Accuracy

Higher Intensity Resolution → More Gray Levels → Better Contrast → Improved Interpretation → Better Computer Vision Performance

5. Detailed Explanation

Spatial resolution specifies the number of pixels used to represent an image, while intensity resolution specifies the number of brightness levels assigned to each pixel. Higher spatial resolution captures finer details but increases storage and computational requirements. Higher intensity resolution improves tonal accuracy and reduces banding, making subtle brightness differences easier to observe. Both characteristics influence preprocessing, feature extraction and deep learning performance. Computer Vision systems balance these parameters to achieve reliable recognition while maintaining computational efficiency. Applications such as medical imaging, autonomous driving, remote sensing and industrial inspection depend on appropriate combinations of spatial and intensity resolution.

6. Applications

- Medical imaging
- Autonomous driving
- Satellite imaging
- Facial recognition
- Industrial inspection
- Remote sensing
- Scientific imaging
- Smart surveillance
- Agriculture
- Mobile photography

7. Conclusion

Spatial and intensity resolution jointly determine digital image quality and directly affect the usability and performance of Computer Vision systems. Selecting appropriate values improves recognition accuracy while balancing storage and computational costs.

8. IEEE References

1. R. C. Gonzalez and R. E. Woods, Digital Image Processing, 4th ed., Pearson, 2018.
2. R. Szeliski, Computer Vision: Algorithms and Applications, 2nd ed., Springer, 2022.
3. W. K. Pratt, Digital Image Processing.
4. D. Forsyth and J. Ponce, Computer Vision: A Modern Approach.
5. I. Goodfellow et al., Deep Learning, MIT Press, 2016.
6. G. Bradski and A. Kaehler, Learning OpenCV.
7. OpenCV Documentation.
8. Y. LeCun et al., Nature, 2015.
9. D. Lowe, IJCV, 2004.
10. J. Canny, IEEE TPAMI, 1986.